

MAHMOUD ELFIL

Junior Software Engineer | Backend, Full-Stack, Security Software & Systems

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PROFESSIONAL SUMMARY

Berlin-based junior software engineer completing a B.Sc. in Networking Engineering in July 2026. Industry experience across Python/FastAPI backend services, React/Next.js dashboards, Python/Qt desktop tooling, Kotlin Android security workflows, Swift iOS capture workflows, Wazuh/WireGuard integrations, and computer-vision preprocessing. Strong fit for backend, full-stack, security software, network tooling, and internal systems roles.

CORE STRENGTHS

Backend / full-stack: FastAPI services, REST APIs, authentication flows, customer dashboards, React/Next.js/TypeScript frontends, Node/Express services, PostgreSQL/MongoDB-backed applications

Security software / network tooling: Wazuh agent deployment, log ingestion, vulnerability-detection workflows, WireGuard VPN integration, Wi-Fi/app risk scoring, evil-twin detection, endpoint visibility

Systems / desktop / mobile: Python/Qt desktop apps, Windows/Linux packaging, Inno Setup installers, Windows services, Android/Kotlin modules, iOS/Swift workflows

Data / CV / algorithms: YOLOv7/YOLOv9 preprocessing, OpenCV, CP-SAT optimization, timetable solving, statistical anomaly detection, low-level CPU/AES projects

EXPERIENCE

Resistine GmbH | Berlin, Germany | Dec 2025 - Present

Desktop and Android Developer, Network Security

- Built a cross-platform Python/Qt desktop tool that automates Wazuh agent installation, configuration, service setup, and reporting to the Wazuh server.
- Replaced a manual onboarding flow involving website setup, generated curl commands, terminal execution, and manual Windows service startup through services.msc.
- Packaged and deployed Windows/Linux builds, including Inno Setup installer scripts, bundled dependencies, configuration files, and system-service setup.
- Deployed across 5-20 customer businesses and supported daily internal use by approximately 20-30 team members.
- Developed customer-facing Android modules in Kotlin for Wi-Fi scanning, Wi-Fi risk scoring, evil-twin detection, installed-app risk scoring, and security reporting.
- Implemented app risk analysis based on permissions, outdated packages, vulnerability signals, and automated Wazuh reporting/log ingestion.
- Integrated Android clients with WireGuard VPN tunnels and server-connected Wazuh reporting flows; owned VPN, Wi-Fi scanning, app scanning, bug fixes, and deployment work.

IAV GmbH | Berlin, Germany | Oct 2024 - Nov 2025

Software Engineering Working Student (iOS, Web, Backend, Computer Vision)

- Developed two Python/FastAPI services covering image-upload workflows, dashboard data delivery, and customer authentication for software-solution workflows.
- Built approximately 20-50 React/Next.js dashboard pages and reusable UI components for customer engineers and internal managers.
- Developed and maintained a Swift iOS app for accessibility-focused image capture, quality filtering, and preprocessing workflows.
- Improved capture reliability with pre-upload image-quality checks, retake flows for noisy images, and video-based best-frame selection for more informative submissions.
- Worked with camera APIs, WebRTC-based connection/authentication flows, and custom preprocessing that used gyroscope data to infer additional depth context.
- Maintained and upgraded YOLOv7/YOLOv9 preprocessing pipelines for datasets with thousands of images and approximately 20-30 GB of automotive/accessibility-focused data.
- Worked with GitLab CI/CD, GitHub Actions, and test automation across software engineering workflows.

SELECTED PROJECTS

University Timetabling Scheduler - Python, OR-Tools CP-SAT, PyQt6, DOCX/CSV/ICS/PDF exporters, pytest, Inno Setup | [GitHub](#)

- Built an independent 12-week university timetabling solver with CP-SAT feasibility solving, optional local-search improvement, save/load snapshots, imports, and configurable hard/soft constraints.
- Models constraints including room conflicts, staff/group overlaps, staff availability, week-specific rules, lab restrictions, preferred times, room capacity, staff daily/weekly limits, and student workload limits.
- Ships with PyQt6 desktop UI, CLI/headless execution, pytest checks, DOCX/CSV/ICS/PDF exporters, and Windows installer packaging; used by German International University for timetable generation.

TypeShift Station - React, Next.js, TypeScript, Express, PostgreSQL, JWT, Zod | [GitHub](#)

- Built a typing-game platform with multiple modes, custom dictionaries, keyboard-first accessibility, local leaderboard caching, authenticated score submission, and offline/PWA behavior.
- Implemented PostgreSQL-backed sessions/scores, leaderboard APIs, real-time multiplayer race rooms, tournament brackets, daily challenge endpoints, and season leaderboard endpoints.
- Added security hardening with JWT session tokens, parameterized SQL, Zod validation, rate limiting, Helmet, origin checks, strict JSON mutation bodies, anti-cheat sanity checks, hardened security headers, backup/restore scripts, and production DB fail-fast behavior.

Anubis ARG Platform - Next.js, React, TypeScript, MongoDB, Cloudinary, Vitest, GitHub Actions | [GitHub](#)

- Built a full-stack ARG platform using Next.js app router, React/TypeScript, MongoDB via the official driver, Cloudinary avatar storage, and a custom hashed-cookie session system.
- Structured ten hand-authored puzzle levels as JSON/MDX/assets with metadata and hints; added unit/component tests with Vitest and React Testing Library plus GitHub Actions for lint, test, and build workflows.

Doompedia - Kotlin/Jetpack Compose, Swift/SwiftUI, data pipeline, offline-first architecture | [GitHub](#)

- Built an offline-first Wikipedia discovery app architecture with native Android and iOS clients, shared schema/ranking/manifest contracts, local caching, bookmarks, history, settings, and title search.
- Designed pack-update tooling for a 1,000,000-card English Wikipedia summary pack, resumable shard downloads, checksum validation, delta-first updates, and fallback to full shard refresh.

Gems Minecraft Mod - Java, Fabric, Gradle, unit tests, Fabric GameTest | [GitHub](#)

- Built a Java/Fabric Minecraft mod adding holder-dependent gem powers, server-owner documentation, balance/config references, in-game configuration flows, and release builds through Gradle.
- Added unit tests and end-to-end Fabric GameTest coverage for integration-style validation.

AES-128 in Assembly - Assembly, cryptography, Rijndael AES | [GitHub](#)

- Implemented Rijndael AES-128 encryption with key expansion and full round operations: SubBytes, ShiftRows, MixColumns, and AddRoundKey.
- Covered block cipher mechanics, substitution-permutation networks, S-box/Rcon usage, key scheduling, and byte-level bitwise operations.

Processor Simulation - C, custom ISA, pipelined CPU simulation | [GitHub](#)

- Implemented a custom processor simulation covering instruction execution, register/memory state, ALU/control logic, and a five-stage pipeline model.
- Handled pipeline hazards and branch-prediction edge cases across loop-heavy instruction sequences in a computer-architecture project.

DataAnalysis - Python, pandas, NumPy, scikit-learn, statistical modeling | [GitHub](#)

- Built an anomaly-detection pipeline with exploratory data analysis, statistical profiling, missing/infinite value detection, Z-score filtering, PDF/PMF modeling, conditional distributions, and performance plots.
- Implemented a custom Naive Bayes classifier with log-space optimization and compared results against scikit-learn GaussianNB, MultinomialNB, and BernoulliNB models.

Portfolio Site - Next.js, Chakra UI, Framer Motion, Three.js, WebGL shaders, Cloudflare deployment | [GitHub](#) | [Site](#)

- Built a responsive personal portfolio with Next.js App Router, Chakra UI, Framer Motion animations, scroll effects, expandable project cards, visual skill tree, and a WebGL/Three.js hero banner.
- Optimized accessibility, responsiveness, animations, and Core Web Vitals; measured local results include 1.06s LCP, 0 CLS, and 176ms INP, deployed on a custom domain through Cloudflare.

TCP Chat Application - Python, sockets, threading, TCP/IP | [GitHub](#)

- Built a Python client-server chat application using TCP sockets and threading for multi-client support, real-time console logging, bidirectional messaging, and graceful disconnect handling.
- Tested with multiple concurrent clients as a networking-focused project.

EDUCATION

German International University, Berlin | 2023 - 2026 | Expected completion: July 2026

B.Sc. Networking Engineering | GPA: 1.09 / 1.0 (A)

- High-relevance coursework: Data Structures and Algorithms, Computer Programming Lab, Computer Organization and System Programming, Communication Networks, Internet, Communication Theory for Engineering, Random Signals and Noise, Discrete Math and Numerical Methods, Signal and System Theory, Probability and Statistics, Systems and Control, Electromagnetics.
- Selected top grades include Data Structures and Algorithms (0.7/A+), Computer Programming Lab (0.7/A+), Communication Theory for Engineering (0.7/A+), Signal and Network Control (1.0/A), Systems and Control (0.7/A+), and Electrical Circuits I/II (0.7/A+).

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Kotlin, Swift, Java, C, C#, Assembly

Backend / Web: FastAPI, Node.js, Express, REST APIs, PostgreSQL, MongoDB, authentication/session flows, React, Next.js, dashboards

Mobile / Desktop: Android/Kotlin, iOS/Swift, SwiftUI/UIKit, Python/Qt, PyQt6, Windows/Linux tooling, Inno Setup

Security / Networking: Wazuh, WireGuard, VPN integration, TCP/IP, Wireshark, Nmap, Wi-Fi scanning, app-permission analysis, vulnerability-detection workflows, endpoint visibility

Data / Computer Vision: YOLOv7, YOLOv9, OpenCV, Pandas, NumPy, statistical modeling, anomaly detection

DevOps / Testing: Docker, Linux, GitLab CI/CD, GitHub Actions, pytest, Vitest, React Testing Library, Gradle, Fabric GameTest

Spoken languages: English (fluent), Arabic (fluent), German (conversational)